

the name and password fields.

2. If the name and password match the already stored values, the user should be able to navigate to the 'purchase' screen.

Required logic for the purchase screen

This is one of the complex screens, and involves multiple actions and validation points. This is the way the logic should flow:

1. Once the screen is opened, the database should be called to fetch the user name, which will be displayed at the top.
2. The database should be called for getting the user credit balance for the application.
3. The user should be able to select any product from the listed products.
4. Upon selecting a product, its name and price should be displayed to confirm the selection.
5. Upon clicking the *Purchase* button, the user's balance should be deducted.
6. Upon clicking the 'Add Balance' button, the user should be able to add a sum to the user application account.

Packaging and testing

To test the app, you need to get it on your phone. First, you have to download the application to your computer and then move it to your phone via Bluetooth or USB cable. I'll tell you how to download it.

1. On the top row, click on the 'Build' button. It will show you the option to download the apk to your computer.
2. After a successful download, the application will be placed in the *Download* folder of your directory or the preferred location you have set for your downloading.
3. You now need to get this apk file to your mobile phone either via Bluetooth or via USB cable. Once you have placed the apk

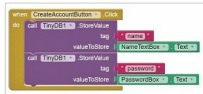


Figure 2: Block editor for creating an account

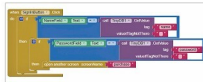


Figure 3: Block editor for 'Sign In' button

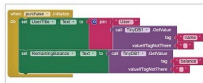


Figure 4: Block editor for purchase



Figure 5: Block editor product list



Figure 6: Block editor 'purchase' button



Figure 7: Block editor 'Add Balance' button



Figure 8: Block editor notifier

file on your SD card, you need to install it. Follow the on-screen instructions to install it. You might get some notification or warning saying that the install is from an untrusted source. Allow this from

the settings, and after successful installation you will see the icon of your application in the menu of your mobile. Here, you will see the default icon, which can be changed and we will tell you how to do this as we move ahead.

I hope your application is working exactly as per the requirements you have given. Now, depending upon your usability and customisation, you can change various things like image, sound and behaviour.

Debugging the application

We have just created the prototype of the application with very basic functionality, but what else might the user be interested in? Let us analyse various cases, which require serious planning, so that using the app does not annoy the user. Consider the following cases:

1. If a user is registered, can we make him/her login automatically to the application?
2. How will we enable lower case and upper case credential validations?
3. What if the price of the selected product is more than the remaining balance — how will we prompt the user to add balance before proceeding for payment?
4. How will a purchase receipt be generated after a successful purchase?
5. What happens if the user gives a text input on the 'add balance' screen?

These are some of the scenarios that might occur and users will be pretty happy seeing them addressed.

Think over all these scenarios and look at how you can integrate these into the application. Do get in touch with me if you fail to accomplish any of the above cases. 

By: Meghraj Singh Beniwal

The author is a B. Tech in electronics and communication, a freelance writer and an Android app developer. He is currently working as an automation engineer at Infosys, Pune. He can be contacted at meghraj.singh01@rediffmail.com or meghrajwithandroid@gmail.com